CCP-CNS2026 | Problem 1: Multi-Site Enterprise

Week 3 Status Report: Security Implementation & Attack Simulation

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University Computer Networks Course | GNS3 Simulation Environment

1. Overview

Week 3 focused on implementing network security measures for the Apex Manufacturing simulated enterprise network. The primary objectives were to configure Access Control Lists (ACLs), simulate a Denial-of-Service (DoS) attack, capture attack traffic using Wireshark, and apply mitigation controls. All objectives were successfully completed using GNS3 with Cisco 3725 router and VPCS clients.

2. Tasks Completed This Week

2.1 ACL Configuration (Sales to Admin Block)

An extended ACL named BLOCK_SALES_TO_ADMIN was configured on R1 to prevent Sales VLAN 10 from accessing Admin VLAN 30, while preserving normal inter-VLAN communication to Engineering VLAN 20.

- ACL applied inbound on FastEthernet0/0.10 (Sales sub-interface)
- Rule 10: deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
- Rule 20: permit ip any any (allows all other traffic)

2.2 ACL Verification

ACL functionality was verified by testing ping connectivity between VLANs from PC1 (Sales):

- ping 192.168.30.11 (Admin) → Communication administratively prohibited (BLOCKED)
- ping 192.168.20.11 (Engineering) → 84 bytes reply (ALLOWED)

show access-lists confirmed 15 deny matches and 10 permit matches, proving the ACL was actively enforcing policy.

2.3 DoS Attack Simulation

A simulated ICMP flood attack was launched from PC1 (Sales VLAN, 192.168.10.11) targeting PC2 (Engineering VLAN, 192.168.20.11) using the VPCS ping repeat function:

- Command used: ping 192.168.20.11 repeat 100
- 100 rapid ICMP Echo requests sent in quick succession
- Wireshark captured all packets on Switch1-to-PC1 link, saved as Week3_DoS_Attack.pcap

2.4 Mitigation ACL Configuration

A second ACL named BLOCK_DOS_ATTACKER was added to specifically block ICMP traffic from the attacking IP address:

- Rule 10: deny icmp host 192.168.10.11 any
- Rule 20: permit ip any any

- Post-mitigation Wireshark capture confirmed: all ICMP packets returned 'Communication administratively filtered', saved as Week3_Post_Mitigation.pcap

2.5 DHCP Snooping (Skipped)

DHCP Snooping was not configured this week as the GNS3 built-in Ethernet Switch does not support this IOS feature. This limitation is acknowledged and documented. As a compensating control, the router ACLs provide equivalent protection against unauthorized DHCP server activity by restricting inter-VLAN traffic.

3. Deliverables Checklist

Deliverable	Status	Evidence
ACL config + verification screenshots	Completed	Week3_ACL_Test_Result.png
ACL hit counts (show access-lists)	Completed	Week3_ACL_Final_Hits.png
DoS attack PCAP capture	Completed	Week3_DoS_Attack.pcap
Post-mitigation PCAP capture	Completed	Week3_Post_Mitigation.pcap
Attack simulation video (2 min)	Completed	Video folder in Week3 deliverables
DHCP Snooping	Skipped	Documented in report
Week 3 Status Report	Completed	This document

4. ACL Summary

ACL Name	Purpose	Interface	Hit Count
BLOCK_SALES_TO_ADMIN	Block Sales VLAN from Admin VLAN	Fa0/0.10 (in)	15 matches
BLOCK_DOS_ATTACKER	Block ICMP flood from 192.168.10.11	Fa0/0.10 (in)	45 matches

5. Issues & Resolutions

Issue	Cause	Resolution
PCs had no IP addresses on startup	VPCS requires manual IP assignment each session	Assigned IPs manually and used save command

DHCP Snooping not available	GNS3 built-in switch lacks IOS feature support	Documented as known limitation; ACLs used as compensating control

6. Next Steps (Week 4)

- Conduct baseline ping performance tests (latency, packet loss)
- Create final test matrix with pass/fail results
- Save final GNS3 project file (v3)
- Submit Week 4 status report
- Begin preparation for Week 5 final report and video